



AQ3.3: Activity Questions 3 - Not Graded



This assignment will not be graded and is only for practice.

Level 1

1 point

Which of the following vectors can be written as the linear combination of vectors $(2, 3)$ and $(1, 2)$ in R^2 with usual addition and scalar multiplication?

☐

$(1, 0)$

☐

$(0, 1)$

☐

$(0, 0)$

☐

Any vector in R^2

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(1, 0)$

$(0, 1)$

$(0, 0)$

Any vector in R^2

1 point

Suppose the set $\{(3, 7, 4), v\}$ is a linearly dependent set in the vector space R^3 with usual addition and scalar multiplication. Which of the following vectors are possible as a candidate for v ?

☐

$(0, 0, 0)$

☐

$(5, \frac{35}{3}, \frac{20}{3})$

☐

$(1, 0, 0)$

☐

$(6, 14, -8)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(0, 0, 0)$

$(5, \frac{35}{3}, \frac{20}{3})$

1 point

Suppose the set $\{(2, 3, 0), (0, 1, 0), v\}$ is a linearly dependent set in the vector space R^3 with usual addition and scalar multiplication. Which of the following vectors are possible as a candidate for v ?

☐

$(2, 3, 1)$

☐

$(1, 2, 0)$

☐


(1, 3, 0)(1, 3, 0)

☐

(0, 1, 1)(0, 1, 1)

☐

$(\pi, e, 0)(\pi, e, 0)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

(1, 2, 0)(1, 2, 0)

(1, 3, 0)(1, 3, 0)

$(\pi, e, 0)(\pi, e, 0)$

If $(3, 4) = \alpha(1, 2) + \beta(1, 3)$ $(3, 4) = \alpha(1, 2) + \beta(1, 3)$ in the vector space R^2 with usual addition and scalar multiplication, then find the value of $\alpha + 3\beta$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -1

1 point

If $(9, 3, 1)$ is a linear combination of the vectors $(1, 1, 1)$, $(1, -1, 1)$ and (x, y, z) $(1, 1, 1)$,

$(1, -1, 1)$ and (x, y, z) as $2(1, 1, 1) + 3(1, -1, 1) + 4(x, y, z) = (9, 3, 1)$

$2(1, 1, 1) + 3(1, -1, 1) + 4(x, y, z) = (9, 3, 1)$, then find the value of $x + y + 2z$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

1 point

Let V be a vector space and v_1, v_2, v_3 and $v_4 \in V$. If v_1 is a linear combination of $v_i, i = 2, 3, 4$ i.e. $v_1 = av_2 + bv_3 + cv_4$ then

☐

v_2 is a linear combination of $v_i, i = 1, 3, 4$.

☐

v_3 is a linear combination of $v_i, i = 1, 4$.

☐

v_2 is a linear combination of v_2 .

☐

v_4 is a linear combination of v_4 .

No, the answer is incorrect.

Score: 0

Accepted Answers:

v_2 is a linear combination of v_2 .

v_4 is a linear combination of v_4 .

1 point

Consider three vectors $v_1 = (1, 0, 0)$, $v_2 = (1, 1, 0)$, $v_3 = (1, 1, 1)$ $v_1 = (1, 0, 0)$, $v_2 = (1, 1, 0)$, $v_3 =$

$(1, 1, 1)$ in the vector space R^3 , with usual addition and scalar multiplication. Which of the following sets is (are) true?

☐

$\{v_1, v_2, v_3\}$ is a linearly dependent set.

☐

$\{v_1 + v_2, v_1 + v_3, v_3\}$ is a linearly dependent set.

☐

$\{v_2 - v_1, v_1, v_2\}$ is a linearly dependent set.

☐

$\{v_1, v_2 - v_1, v_3 - v_2, v_3\}$ is a linearly dependent set.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{v_2 - v_1, v_1, v_2\}$ is a linearly dependent set.

$\{v_1, v_2 - v_1, v_3 - v_2, v_3\}$ is a linearly dependent set.

Level - 2

1 point

Let S be a subset of R^3 which is linearly dependent. Which of the following options are true?

☐

$S \cup \{(1, 0, 0)\}$ must be linearly dependent.

☐

$S \cup \{v\}$ must be linearly dependent for any $v \in R^3$.

☐

$S \setminus \{v\}$ must be linearly dependent for any $v \in R^3$.

☐

There may exist some $v \in S$, such that $S \setminus \{v\}$ is still linearly dependent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$S \cup \{(1, 0, 0)\}$ must be linearly dependent.

$S \cup \{v\}$ must be linearly dependent for any $v \in R^3$.

There may exist some $v \in S$, such that $S \setminus \{v\}$ is still linearly dependent.

$M_{3 \times 3}(R)$ denotes the vector space consisting of real square matrices of order 3 with usual matrix addition and scalar multiplication. Consider the following matrices:

$$M_1 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}, M_2 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}, M_3 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & 2 & 1 \end{bmatrix}, M_4 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 1 & -1 & 1 \end{bmatrix}$$

$$M_1 = \begin{bmatrix} 1 & 0 & 1 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}, M_2 = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}, M_3$$

$$= \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 2 & 1 \\ 1 & 0 & 1 & 0 & 1 & 2 & 1 \\ 1 & 0 & 1 & 0 & 1 & 2 & 1 \end{bmatrix}, M_4 = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & -1 & 1 \\ 1 & 0 & 1 & 0 & 1 & -1 & 1 \\ 1 & 0 & 1 & 0 & 1 & -1 & 1 \end{bmatrix}$$

Use the above information to answer questions 9 and 10.

1 point

Which of the following options are correct?

☐

$M_1 - M_2 = 0$ where 0 denotes the zero matrix of order 3.

☐

$$2M_1 + M_2 = M_3 \quad 2M_1 + M_2 = M_3$$

☐

$$2M_1 - M_2 - M_3 = 0 \quad 2M_1 - M_2 - M_3 = 0, \text{ where } 0 \text{ denotes the zero matrix of order } 3.$$

☐

$$2M_2 - M_1 - M_4 = 0 \quad 2M_2 - M_1 - M_4 = 0, \text{ where } 0 \text{ denotes the zero matrix of order } 3.$$

☐

The tuple (a, b, c) is unique for which $aM_1 + bM_3 + cM_4 = 0$ holds, where 0 denotes the zero matrix of order 3.

☐

The pair (a, b) is unique for which $aM_1 + bM_2 = 0$ holds, where 0 denotes the zero matrix of order 3.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$2M_1 - M_2 - M_3 = 0 \quad 2M_1 - M_2 - M_3 = 0, \text{ where } 0 \text{ denotes the zero matrix of order } 3.$$

$$2M_2 - M_1 - M_4 = 0 \quad 2M_2 - M_1 - M_4 = 0, \text{ where } 0 \text{ denotes the zero matrix of order } 3.$$

The pair (a, b) is unique for which $aM_1 + bM_2 = 0$ holds, where 0 denotes the zero matrix of order 3.

1 point

Which of the following options is(are) true?

☐

$\{M_1, M_2\}$ is a linearly dependent set.

☐

$\{M_1, M_2, M_3\}$ is a linearly dependent set.

☐

$\{M_1, M_2, M_4\}$ is a linearly dependent set.

☐

$\{M_1, M_3, M_4\}$ is a linearly dependent set.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{M_1, M_2, M_3\}$ is a linearly dependent set.

$\{M_1, M_2, M_4\}$ is a linearly dependent set.

$\{M_1, M_3, M_4\}$ is a linearly dependent set.

[Check Answers](#)

Your score is: 0/10